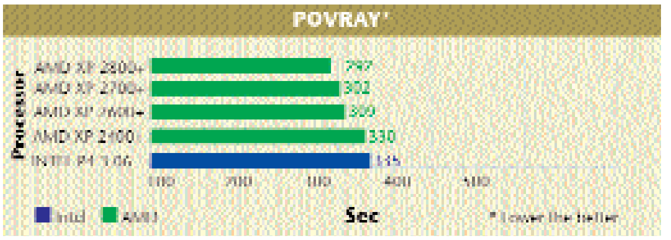


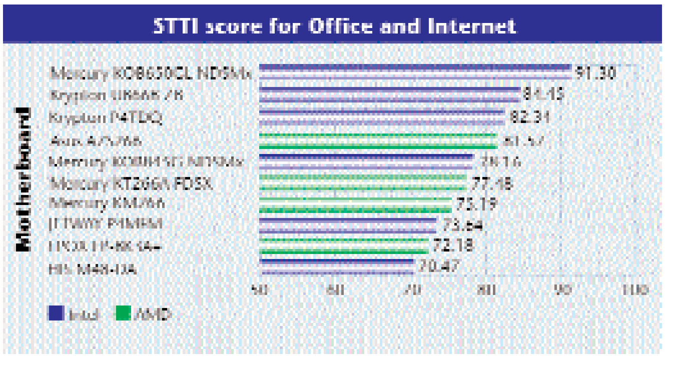
The Celeron continued to suffer in our benchmarks and even its speed couldn't save it. In the POVRAY benchmark, the Celeron took a painstaking 709 seconds (11 minutes 49 seconds) to complete the test. In the Specview Perf test too, it was last on the charts with a low score of 75.22 fps. Although performance was admirable, price played havoc with the Intel processors and caused their STTI scores to drop: the P4 3.06 GHz is a great performer by all means but its price tag puts it at number 5 (74.93 STTI), whereas the 2.8 GHz Pentium 4 is two notches further down at position 8 with a score of 71.63 STTI.

Our recommended solution

As far as content creation goes, nothing beats the MSI KT4-Ultra with an AMD XP 1700+ processor, the reason being extremely competitive pricing of both the motherboard and the processor. The MSI supports up to 3 GB of DDR memory. The AGP slot is the new and improved AGP 8x standard complaint, offering plenty of bandwidth for those 3D applications. The board also features onboard FireWire, Ethernet, USB 2.0, RAID and Bluetooth—heaven-sent for this application area. A total of six PCI slots leave plenty of room for future upgrades. The Athlon XP 1700+ processor is, of course, a great performer at its price point, posting healthy scores of 76.4 under Content Creation 2001 and 4406 under PCMark 2002 Pro.

Office and Internet

Office and Internet applications such as Microsoft Word, Excel, Macromedia Dreamweaver and Internet Explorer have become a part of our daily life. To test a CPU's performance under applications such as Word, Excel, Web authoring tools, etc, we used the Ziff Davis Business Creation benchmark that executes a script that runs all these programs in turn and then logs the score. To test the performance of applications such as WinZip, Macromedia Dreamweaver, etc we used the Ziff Davis Content Creation benchmark. PCMark 2002 Pro, a synthetic test, is more application specific, that is, the benchmarking applications are basically scripts written to perform a series of tasks and then log the resultant score. Hence, the way it uses the CPU instruction, or the OS, will directly be reflected in the score. For running office and Internet applications, one does not really need the fastest processor in the world. Here anything above a core speed of 1 GHz will do fine even for today's most demanding applications, if coupled with at least 128 MB of memory.



In Memory of Standards

With the arrival of various memory standards, choosing a motherboard has become a more complicated and confusing task. At present, the market offers three different varieties of memory modules: SDRAM, DDR SDRAM and RDRAM. The problem has been compounded with the different incarnations of one standard itself. As an example, DDR SDRAM (also known as DDR RAM) is available as DDR266, DDR333, DDR400 and the latest, DDR433—the numbers denote the operating frequency of the memory module. Similarly, RDRAM also has many variants to choose from. The million-rupee question is, which memory standard should one adopt for optimum performance?

If you venture into the market for buying a memory module, more often than not you'll end up buying an SDRAM (Synchronous Dynamic Random Access Memory) stick, simply because it is the cheapest of the available standards and has been in use for quite some time now. Nearly all motherboards have chipsets that support this memory standard. The fastest variant is the PC133, where the memory runs at 133 MHz. SDRAM makes sense for those of you who wish to use their PC for office applications and Internet surfing. Keep in mind that this standard is on the verge of extinction and it could be out of the market within six months.

AMD adopted the DDR (Double Data Rate) SDRAM standard to counter Intel's RDRAM (RAMBUS Dynamic Random Access Memory) that had been introduced by Intel for its Pentium 4. As of today, DDR SDRAM is fast gaining a reputation as the most favoured and the most dominating of all memory standards. It's just a matter of time before it replaces SDRAM even in the low-end boards. Its success can be attributed to the performance gain, the falling prices and the strong support it enjoys from nearly all motherboard and chipset vendors.

DDR SDRAM has one significant point of difference from the SDRAM standard—data is transferred twice as quickly as its older cousin, hence the name. Moreover, with the introduction of new high-speed DDR memory standards such as DDR400 and DDR433, and the subsequent introduction of chipsets such as VIA's KT400 which support these standards, the performance gain that can be achieved is significant. The price factor is somewhat of a deterrent, with even the lowest speed DDR200 sticks costing almost twice as much as a PC133 module. Moreover, DDR400 and DDR433 modules aren't even available in the Indian market. Another sore point is the compatibility issues. Certain manufacturers mention in their manual the list of all memory manufacturers that the chipset supports.

RDRAM is still the fastest available memory for the Pentium 4. But its price has led to a low level of acceptance. With the new high-speed DDR standards such as DDR433 that promise performance levels on par with that of RDRAM, the performance advantage that RDRAM had also seems to have diminished. This could hammer the final nail into its coffin.